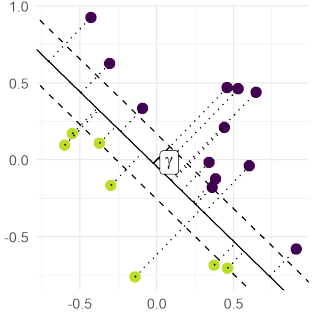
# Introduction to Machine Learning

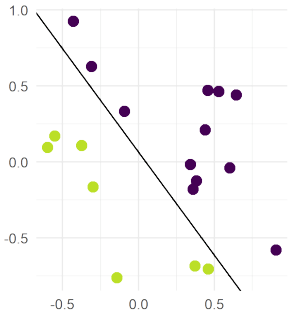
## Linear Support Vector Machines
## Linear Hard Margin SVM

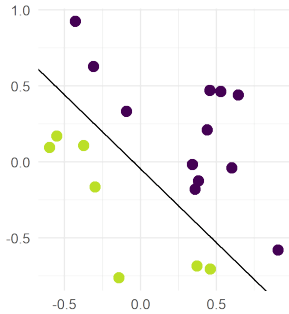


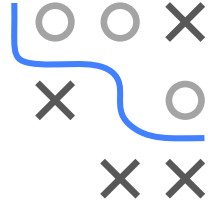

**Learning goals**

- Know that the hard-margin SVM maximizes the margin between data points and hyperplane
- Know that this is a quadratic program
- Know that support vectors are the data points closest to the separating hyperplane

# LINEAR CLASSIFIERS

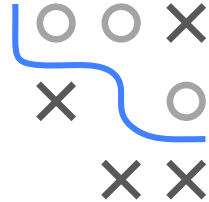


- We want study how to build a binary, linear classifier from solid geometrical principles.
- Which of these two classifiers is “better”?

# SUPPORT VECTOR MACHINES: GEOMETRY



For labeled data $\mathcal{D} = \left(\left(\mathbf{x}^{(1)}, y^{(1)}\right), \ldots, \left(\mathbf{x}^{(n)}, y^{(n)}\right)\right)$, with $y^{(i)} \in \{-1, +1\}$:

- Assume linear separation by $f(\mathbf{x}) = \boldsymbol{\theta}^\top \mathbf{x} + \theta_0$, such that all $+$-observations are in the positive halfspace

$$\{\mathbf{x} \in \mathcal{X} : f(\mathbf{x}) > 0\}$$

  and all $-$-observations are in the negative halfspace

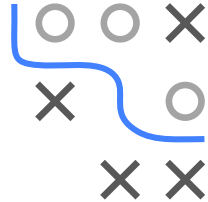
$$\{\mathbf{x} \in \mathcal{X} : f(\mathbf{x}) < 0\}.$$

- For a linear separating hyperplane, we have

$$y^{(i)} \underbrace{\left(\boldsymbol{\theta}^\top \mathbf{x}^{(i)} + \theta_0\right)}_{=f\left(\mathbf{x}^{(i)}\right)} > 0 \quad \forall i \in \{1, 2, \ldots, n\}.$$

# MAXIMUM MARGIN SEPARATION

We formulate the desired property of a large “safety margin” as an optimization problem:

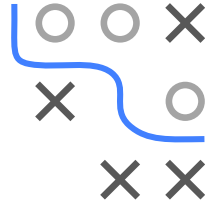


$$\max_{\boldsymbol{\theta},\theta_0} \quad \gamma$$
$$\text{s.t.} \quad d\left(f, \mathbf{x}^{(i)}\right) \geq \gamma \quad \forall\, i \in \{1, \ldots, n\}.$$

- The constraints mean: We require that any instance $i$ should have a “safety” distance of at least $\gamma$ from the decision boundary defined by $f(= \boldsymbol{\theta}^T \mathbf{x} + \theta_0) = 0$.
- Our objective is to maximize the “safety” distance.

# MAXIMUM MARGIN SEPARATION

We reformulate the problem:

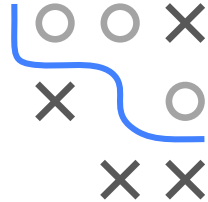


$$\max_{\boldsymbol{\theta},\theta_0} \quad \gamma$$
$$\text{s.t.} \quad \frac{y^{(i)}\left(\left\langle \boldsymbol{\theta}, \mathbf{x}^{(i)} \right\rangle + \theta_0\right)}{\|\boldsymbol{\theta}\|} \geq \gamma \quad \forall\, i \in \{1, \ldots, n\}.$$

- The inequality is rearranged by multiplying both sides with $\|\boldsymbol{\theta}\|$:

$$\max_{\boldsymbol{\theta},\theta_0} \quad \gamma$$
$$\text{s.t.} \quad y^{(i)}\left(\left\langle \boldsymbol{\theta}, \mathbf{x}^{(i)} \right\rangle + \theta_0\right) \geq \|\boldsymbol{\theta}\|\gamma \quad \forall\, i \in \{1, \ldots, n\}.$$

# QUADRATIC PROGRAM



We derived the following optimization problem:

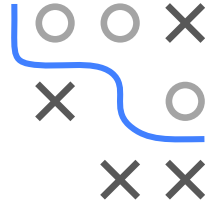
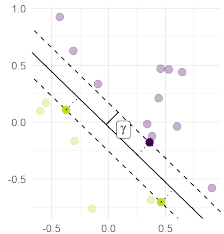
$$\begin{aligned} \min_{\boldsymbol{\theta},\theta_0} \quad & \frac{1}{2}\|\boldsymbol{\theta}\|^2 \\ \text{s.t.} \quad & y^{(i)}\left(\left\langle \boldsymbol{\theta}, \mathbf{x}^{(i)} \right\rangle + \theta_0\right) \geq 1 \quad \forall\, i \in \{1, \ldots, n\}. \end{aligned}$$

This turns out to be a **convex optimization problem** – particularly, a **quadratic program**: The objective function is quadratic, and the constraints are linear inequalities.

This is called the **primal** problem. We will later show that we can also derive a dual problem from it.

We will call this the **linear hard-margin SVM**.

# SUPPORT VECTORS



- There exist instances $(\mathbf{x}^{(i)}, y^{(i)})$ with minimal margin $y^{(i)} f\left(\mathbf{x}^{(i)}\right) = 1$, fulfilling the inequality constraints with equality.
- They are called **support vectors (SVs)**. They are located exactly at a distance of $\gamma = 1/\|\boldsymbol{\theta}\|$ from the separating hyperplane.
- It is already geometrically obvious that the solution does not depend on the non-SVs! We could delete them from the data and would arrive at the same solution.